

Steps in Hypothesis Testing

Step 1: Set-up Null and Alternative Hypothesis: H_0 and H_a .

What we want to prove is in the Alternative Hypothesis

Step 2: Specify level of significance α (e.g., $\alpha = 1\%$ or 5%).

This is the maximum type 1 error allowed!

Step 3: Based on the sampling distribution of the sample statistic,

- Compute the p-value:

How likely do we observe evidence as contradictory to H_0 as the given data, if we assume H_0 is true?

- Make decision: if p-value $\leq \alpha$, then reject H_0 ,

Alternative method for step 3, compute the test-statistic and compare with critical values (SEE NEXT PAGE).

Interpretation of the p-value: How significant is the observed data against H_0 ?

A small p-value indicates that either:

- H_0 is true but we have been very unlucky (we've drawn a misrepresentative sample data);
- or H_0 is wrong.

Because we don't typically expect to be very unlucky, so when p-value is very small we say: the data, all by itself, makes us suspicious about H_0 . **Thus, we reject H_0 and conclude H_a .**

When p-value is relatively large, the data is not strong enough to reject H_0 . **We say we fail to reject H_0 .**

If p-value is

- | | |
|-------------------------|---|
| • Less than 0.01 | Strong evidence to conclude that H_a is true. |
| • Between 0.01 and 0.05 | Quite good evidence to conclude that H_a is true |
| • Between 0.05 and 0.10 | Weak evidence to conclude that H_a is true. |
| • Greater than 0.10 | insufficient evidence to conclude that H_a is true. |

Testing for one population mean

	Known σ	Unknown σ
	z-test statistics $z = \frac{\bar{X} - \mu_0}{\sigma / \sqrt{n}}$	t-test statistic $t = \frac{\bar{X} - \mu_0}{s / \sqrt{n}}$
Upper tailed test $H_0: \mu \leq \mu_0; H_a: \mu > \mu_0$	Reject H_0 if $z \geq z_\alpha$ Reject H_0 If p-value $\leq \alpha$ p-value = $\Pr(Z \geq z)$	Reject H_0 if $t \geq t_{\alpha, (n-1)}$ Reject H_0 If p-value $\leq \alpha$ p-value = $\Pr(T \geq t)$
Lower tailed test $H_0: \mu \geq \mu_0; H_a: \mu < \mu_0$	Reject H_0 if $z \leq -z_\alpha$ Reject H_0 If p-value $\leq \alpha$ p-value = $\Pr(Z \leq z)$	Reject H_0 if $t \leq -t_{\alpha, (n-1)}$ Reject H_0 If p-value $\leq \alpha$ p-value = $\Pr(T \leq t)$
Two Tailed test $H_0: \mu = \mu_0; H_a: \mu \neq \mu_0$	Reject H_0 if either $z \geq z_{\alpha/2}$ or $z \leq -z_{\alpha/2}$ Reject H_0 If p-value $\leq \alpha$ p-value = $2 \times \Pr(Z \geq z)$	Reject H_0 if either $t \geq t_{\alpha/2, (n-1)}$ or $t \leq -t_{\alpha/2, (n-1)}$ Reject H_0 If p-value $\leq \alpha$ p-value = $2 \times \Pr(T \geq t)$

Testing for one population proportion

$$\text{z-test statistics} \quad z = \frac{\bar{p} - p_0}{\sqrt{p_0(1-p_0)} / \sqrt{n}}$$

Upper tailed test $H_0: p \leq p_0; H_a: p > p_0$	Lower tailed Test $H_0: p \geq p_0; H_a: p < p_0$	Two Tailed test $H_0: p = p_0; H_a: p \neq p_0$
Reject H_0 if $z \geq z_\alpha$ p-value = $\Pr(Z \geq z)$	Reject H_0 if $z \leq -z_\alpha$ Reject H_0 If p-value $\leq \alpha$ p-value = $\Pr(Z \leq z)$	Reject H_0 if either $z \geq z_{\alpha/2}$ or $z \leq -z_{\alpha/2}$ Reject H_0 If p-value $\leq \alpha$ p-value = $2 \times \Pr(Z \geq z)$